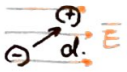
§21 Electric Field

Coulomb's Law $F=\frac{1}{4\pi\varepsilon_0}\frac{|q_1 q_2|}{r^2}$

Electric Field $\bar{E}=\frac{\bar{F}_0}{q}$ (def)

point charge $\bar{E}=\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$

Electric Dipoles



$\bar{\tau}=\bar{p}\times\bar{E}$ $U=-\bar{p}\cdot\bar{E}$

where $\bar{p}=q\bar{d}$

(electric dipole moment)

§22 Gauss's Law

Electric Flux $\Phi_E=\int\bar{E}\cdot d\bar{A}$

Gauss's Law $\Phi_E=\int\bar{E}\cdot d\bar{A}=\frac{Q_{encl}}{\varepsilon_0}$

§23 Electric Potential

Electric Potential Energy

$W_{a\to b}=U_a-U_b$ (for conservative force)

$U=\frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r}$

$U=\frac{q_0}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1}+\frac{q_2}{r_2}+\cdots+\frac{q_n}{r_n}\right)$

Electric Potential $V=\frac{U}{q_0}$ (def)

$V=\frac{1}{4\pi\varepsilon_0}\frac{q}{r}$

$V=\frac{1}{4\pi\varepsilon_0}\sum\frac{q_i}{r_i}$

$V=\frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r}$

$V_a-V_b=\int_a^b\bar{E}\cdot d\bar{l}$

Equipotential Surface

⊥ electric field line

surface of a conductor

interior of a conductor

Potential Gradient

$\bar{E}=-\nabla V=-\left(\frac{\partial V}{\partial x},\frac{\partial V}{\partial y},\frac{\partial V}{\partial z}\right)$

§24 Capacitance

Capacitance $C=\frac{Q}{V_{ab}}$ (def) unit: F

$C=\varepsilon_0\frac{A}{d}$ (vacuum)

in series $\frac{1}{C_{eq}}=\frac{1}{C_1}+\frac{1}{C_2}+\cdots+\frac{1}{C_n}$

in parallel $C_{eq}=C_1+C_2+\cdots+C_n$

Energy in Capacitors

$U=\frac{Q^2}{2C}=\frac{1}{2}CV^2=\frac{1}{2}QV$

$u=\frac{1}{2}\varepsilon_0E^2$ (vacuum)

Dielectrics

dielectric const. $K=\frac{C}{C_0}$ (def)

permitivity $\varepsilon=K\varepsilon_0$ (def)

* For capacitors with dielectric

$C=KC_0=K\varepsilon_0\frac{A}{d}=\varepsilon\frac{A}{d}$

$u=\frac{1}{2}K\varepsilon_0E^2=\frac{1}{2}\varepsilon E^2$

$\oint K\bar{E}\cdot d\bar{A}=\frac{Q_{encl\text{-}free}}{\varepsilon_0}$ Gauss's Law

§25 Current & Resistance

Current & ~ Density

$I=\frac{dQ}{dt}=n|q|v_dA$

$\bar{J}=nq\bar{v}_d$ $J=\frac{I}{A}$

where n - concentration

v_d - drift velocity

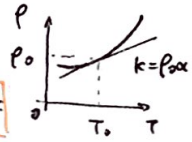
Resistivity $\rho=\frac{E}{J}$ (def)

$\rho(T)=\rho_0[1+\alpha(T-T_0)]$ Ohm's Law

α - tempreture coeff. of ρ

Resistors

$V=IR$ $R=\frac{\rho L}{A}$



Circuits & emf

$V_{ab}=\varepsilon-Ir$

Energy & Power in Circuits

$P=V_{ab}I$ (general)

$P=V_{ab}I=I^2R=\frac{V_{ab}^2}{R}$ (resistor)

§26 DC Circuits

series $R_{eq}=R_1+R_2+\cdots+R_n$

parallel $\frac{1}{R_{eq}}=\frac{1}{R_1}+\frac{1}{R_2}+\cdots+\frac{1}{R_n}$

KCL $\sum I=0$ (at junctions)

KVL $\sum V=0$ (in loop)

R-C Circuits

charging:

$q=C\varepsilon(1-e^{-\frac{t}{RC}})=Q_f(1-e^{-\frac{t}{RC}})$

$i=\frac{dq}{dt}=\frac{\varepsilon}{R}e^{-\frac{t}{RC}}=I_0e^{-\frac{t}{RC}}$

discharging:

$q=Q_0e^{-\frac{t}{RC}}$

$i=\frac{dq}{dt}=-\frac{Q}{RC}e^{-\frac{t}{RC}}=I_0e^{-\frac{t}{RC}}$

§27 Magnetic Field

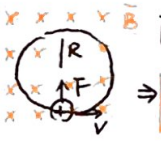
Magnetic Forces $\bar{F}=q\bar{v}\times\bar{B}$

Magnetic Flux $\Phi_B=\int\bar{B}\cdot d\bar{A}$

unit: $1\,Wb=1\,T\cdot m^2$

$\oint\bar{B}\cdot d\bar{A}=0$ Gauss's Law

Uniform Circular Motion



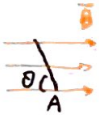
$F=m\frac{v^2}{R}=|q|vB$

$\Rightarrow R=\frac{mv}{|q|B}$

Magnetic Force on Conductors

$\bar{F}=I\bar{l}\times\bar{B}$ $d\bar{F}=Id\bar{l}\times\bar{B}$

Magnetic Torque



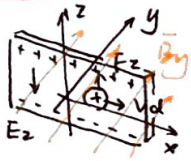
$\bar{\tau}=\bar{\mu}\times\bar{B}$ $U=-\bar{\mu}\cdot\bar{B}$

where $\bar{\mu}=I\bar{A}$

(magnetic dipole moment)

The Hall Effect

$nq=-\frac{J_x B_y}{E_z}$



§28 Source of Mag. Field

Magnetic Field of moving charge $\bar{B}=\frac{\mu_0}{4\pi}\frac{q\bar{v}\times\hat{r}}{r^2}$

wire $d\bar{B}=\frac{\mu_0}{4\pi}\frac{Id\bar{l}\times\hat{r}}{r^2}$

long. straight wire $\bar{B}=\frac{\mu_0 I}{2\pi r}$

Magnetic Force between wires

$\frac{F}{L}=\frac{\mu_0 II'}{2\pi r}$

Ampere's Law

$\oint\bar{B}\cdot d\bar{l}=\mu_0 I_{encl}$

§29 Electromagnetic Induction

Faraday's Law $\varepsilon=-\frac{d\Phi_B}{dt}$

Lenz's Law 楞次定律

Motional emf

$\varepsilon=vBL$

$\varepsilon=\oint(\bar{v}\times\bar{B})\cdot d\bar{l}$ (general)

Induced Electric Fields

$\oint\bar{E}\cdot d\bar{l}=-\frac{d\Phi_B}{dt}$

Maxwell's Equations

$\oint\bar{E}\cdot d\bar{A}=\frac{Q_{encl}}{\varepsilon_0}$ Gauss

$\oint\bar{B}\cdot d\bar{A}=0$ Gauss

$\oint\bar{B}\cdot d\bar{l}=\mu_0(i_{ct}+i_D)_{encl}$ Ampere

$\oint\bar{E}\cdot d\bar{l}=\frac{d\Phi_B}{dt}$ Faraday

where displacement current

$i_D=\varepsilon\frac{d\Phi_E}{dt}$

§30 Inductance

Mutual Inductance

$\varepsilon_2=-M\frac{di_1}{dt}$ $\varepsilon_1=-M\frac{di_2}{dt}$

$M=\frac{N_2\Phi_{B2}}{i_1}=\frac{N_1\Phi_{B1}}{i_2}$

Self-inductance

$\varepsilon=-L\frac{di}{dt}$ $L=\frac{N\Phi_B}{i}$

Mag. Field Energy

$U=\frac{1}{2}LI^2$

$u=\frac{B^2}{2\mu_0}$ (vacuum) $u=\frac{B^2}{2\mu}$

R-L Circuits

$i(t)=i(\infty)+[i(0)-i(\infty)]e^{-\frac{Rt}{L}}$

L-C Circuits

$q=Q\cos(\omega t+\phi)$

$i=-\omega Q\sin(\omega t+\phi)$

where $\omega=\sqrt{\frac{1}{LC}}$

L-R-C Series Circuits

$\frac{d^2q}{dt^2}-\frac{R}{L}\frac{dq}{dt}+\frac{1}{LC}q=0\ (\Delta<0)$

$q=Qe^{-\frac{R}{2L}t}\cos(\omega' t+\phi)$

where $\omega'=\sqrt{\frac{1}{LC}-\frac{R^2}{4L^2}}$

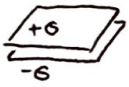
Electric Field $\bar{E}$

$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$

$E = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$

$E = \begin{cases} \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}, & r>R \\ 0, & r<R \end{cases}$

$E = \frac{\sigma}{2\varepsilon_0}$



$E = \frac{\sigma}{\varepsilon_0}$ (⊟)

$E = \frac{\sigma}{\varepsilon_0}$ (⊞)

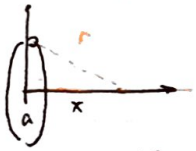
charged conductor

E of an dipole

$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\cdot 2\cos\alpha$

$= \frac{q}{4\pi\varepsilon_0}\frac{l}{r^3}$

E & V of a ring



$dE = \frac{1}{4\pi\varepsilon_0}\frac{dQ}{x^2+a^2}$

$dE_x = dE\cos\alpha = \cdots = \frac{1}{4\pi\varepsilon_0}\frac{\lambda x}{(x^2+a^2)^{\frac{3}{2}}}ds$

$E_x = \int dE_x = \cdots = \frac{1}{4\pi\varepsilon_0}\frac{\lambda x}{(x^2+a^2)^{\frac{3}{2}}}(2\pi a)$

$Q = 2\pi a\lambda$

$\frac{1}{4\pi\varepsilon_0}\frac{Qx}{(x^2+a^2)^{\frac{3}{2}}}$

$V = \frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r} = \cdots = \frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{x^2+a^2}}$

E & V of a line segment

$dE = \frac{1}{4\pi\varepsilon_0}\frac{dQ}{r^2}$

$= \frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\frac{dy}{(x^2+y^2)}$

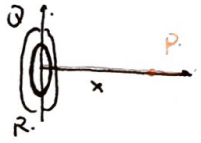
$dE_x = \frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\frac{x\,dy}{(x^2+y^2)^{\frac{3}{2}}}$

$E_x = \int dE_x = \cdots = \frac{Q}{4\pi\varepsilon_0}\frac{1}{x\sqrt{x^2+a^2}}$

$dV = \frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\frac{dy}{\sqrt{x^2+y^2}}$

$V = \int dV = \cdots = \frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\ln\left(\frac{\sqrt{a^2+x^2}+a}{\sqrt{a^2+x^2}-a}\right)$

$\bar{E}$ of a disk

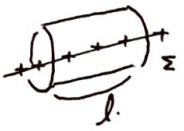


$dE_x = \frac{1}{4\pi\varepsilon_0}\frac{2\pi\sigma r x\,dr}{(x^2+r^2)^{\frac{3}{2}}}$

$E = \int dE_x = \cdots \overset{t=x^2+r^2}{=} \frac{\sigma x}{2\varepsilon_0}\left[1-\frac{1}{\sqrt{(R/x)^2+1}}\right]$

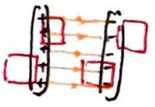
Use Gauss's Law!

E of line charge



E of an infinite plane

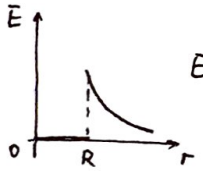
E of parallel plates



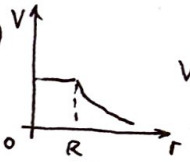
E & V of a charged conducting sphere

$\Phi_E = \int\bar{E}\cdot d\bar{A} = E(r)4\pi r^2$

Gauss's Law ⇒ E



$E(r) = \begin{cases} \frac{Q}{4\pi\varepsilon_0}\frac{1}{r^2}, & r>R \\ 0, & r<R \end{cases}$

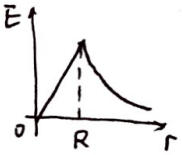


$V(r) = \begin{cases} \frac{Q}{4\pi\varepsilon_0}\frac{1}{r}, & r>R \\ \frac{Q}{4\pi\varepsilon_0}\frac{1}{R}, & r<R \end{cases}$

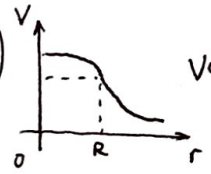
E & V of a uniformly charged sphere

$\Phi_E = \int\bar{E}\cdot dA = E(r)4\pi r^2$

Gauss's Law ⇒ E



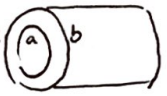
$E(r) = \begin{cases} \frac{Q}{4\pi\varepsilon_0}\frac{1}{r^2}, & r>R \\ \frac{Q}{4\pi\varepsilon_0}\frac{r}{R^3}, & r<R \end{cases}$



$V(r) = \begin{cases} \frac{Q}{4\pi\varepsilon_0}\frac{1}{r}, & r>R \\ \frac{Q}{4\pi\varepsilon_0}\frac{3R^2-r^2}{2R^3}, & r<R \end{cases}$

V of line charge

$V_a - V_b = \int_a^b\bar{E}\cdot dl = \cdots = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_b}{r_a}$

⇒ 

$V_{ab} = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_b}{r_a}$

Magnetic Field $\bar{B}$

$B = \frac{\mu_0 I}{2\pi r}$

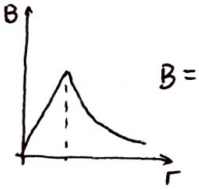
$B = \frac{\mu_0 I a^2}{2(x^2+a^2)^{\frac{3}{2}}}$

When $x=0$ $B = \frac{\mu_0 I}{2a}$

B of long cylindrical conductor

$J = \frac{I}{\pi R^2} \Rightarrow I_{encl} = J\pi r^2$

$= I\frac{r^2}{R^2}$



$B = \begin{cases} \frac{\mu_0 I}{2\pi}\frac{r}{R^2}, & r<R \\ \frac{\mu_0 I}{2\pi r}, & r>R \end{cases}$

B of a solenoid

Ampere's Law

$B = \begin{cases} \mu_0 nI, & r<R \\ \approx 0, & r>R \end{cases}$

n - density

$I_{encl} = nLI$

B of a toroidal solenoid

$B = \begin{cases} \frac{\mu_0 NI}{2\pi r}, & R_1<r<R_2 \\ \approx 0, & r>R_2 \text{ or } r<R_1 \end{cases}$

$I_{encl} = NI$

积分表

$\int k\,dx = kx + C$

$\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C \quad (\alpha\neq-1)$

$\int\frac{1}{x}dx = \ln|x| + C$

$\int e^x dx = e^x + C$

$\int\sin x\,dx = -\cos x + C$

$\int\cos x\,dx = \sin x + C$

$\int\sec^2 x\,dx = \tan x + C$

$\int\csc^2 x\,dx = -\cot x + C$

$\int\sec x\tan x\,dx = \sec x + C$

$\int\csc x\cot x\,dx = -\csc x + C$

$\int\frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C \quad (a>0)$

$\int\frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a} + C \quad (a\neq0)$

$\int\frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C \quad (a\neq0)$

$\int\sec x\,dx = \ln|\sec x + \tan x| + C$

$\int\csc x\,dx = \ln|\csc x - \cot x| + C$

$\int\frac{dx}{\sqrt{x^2+a^2}} = \ln(x+\sqrt{x^2+a^2}) + C \quad (a>0)$

$\int\frac{dx}{\sqrt{x^2-a^2}} = \ln|x+\sqrt{x^2-a^2}| + C \quad (a>0)$

常数

$\varepsilon_0 = 8.85\times10^{-12}$

$\frac{1}{4\pi\varepsilon_0} = 8.988\times10^9$

$\mu_0 = 1.257\times10^{-6}$

$e = 1.602\times10^{-19}$ C

$m_p = 1.673\times10^{-27}$ kg

$m_e = 9.109\times10^{-31}$ kg

Sinda

VP240 Mid